

Python: Banking Transaction Exception

Create a Python program to simulate banking transactions with transaction history, security measures, and error handling features.

Implement two classes:

1. **Transaction Class**

2. **init(self, transaction_type, amount, balance):** Initializes a new transaction

str(self): Returns a string representation of the transaction in the form "Type: {self.transaction_type}, Amount: {self.amount}, Balance: {self.balance}"

BankAccount Class

5. **init(self, initial_balance):** Initializes an account with specified balance
6. **process_transaction(self, transaction_amount, transaction_type):** Processes a transaction

Initial Conditions * Account balance starts at 1500.0

Transaction Types * **deposit:** * Adds the transaction amount to the balance * Updates transaction history * Returns: * "Transaction successful!" * "Updated account balance: {self.balance}" * **withdraw:** * Deducts the transaction amount if sufficient funds are available * Updates transaction history * If funds are insufficient, raises an **InsufficientFundsError** * Returns: * "Transaction successful!" * "Updated account balance: {self.balance}" * **view_transaction_history:** * Displays transaction history, with each transaction shown in the format "Type: {self.transaction_type}, Amount: {self.amount}, Balance: {self.balance}"

Custom Exceptions * **InsufficientFundsError:** * Raised when withdrawal exceeds available funds * Message: "Not enough funds to withdraw {transaction_amount}. Current balance: {balance}" * status_code: 400 * **InvalidTransactionType:** * Raised for invalid transaction types * Message: "Invalid transaction type '{transaction_type}'. Accepted types: deposit, withdraw, view_transaction_history" * status_code: 400

Constraints * All transaction amounts are positive floats in the form of a string. * The account balance is a non-negative float. * $1 \leq n \leq 1000$

Input Format For Custom Testing The first line contains an integer, n , representing the number of transactions. For the next n lines: If the transaction type is not `view_transaction_history`, the line contains the transaction type followed by the transaction amount in the next line. If the transaction type is `view_transaction_history`, the line contains only the transaction type.

Sample Case 0 Sample Input For Custom Testing

```
3
deposit
130000.0
withdraw
3400.0
view_transaction_history
```

Sample Output

```
Transaction successful!
Updated account balance: 131500.0
Transaction successful!
Updated account balance: 128100.0
Type: deposit, Amount: 130000.0, Balance: 131500.0
Type: withdraw, Amount: 3400.0, Balance: 128100.0
```

Explanation First, 130000.0 is deposited, bringing the balance to 131500.0, and then 3400.0 is withdrawn. Finally, the transaction history is viewed.

Sample Case 1 Sample Input For Custom Testing

```
3
deposit
500.0
withdraw
1000.0
invalid_type
```

Sample Output

```
Transaction successful!
Updated account balance: 2000.0
Transaction successful!
Updated account balance: 1000.0
Error (400): Invalid transaction type 'invalid_type'. Accepted types: deposit, withdraw, view_
```

Explanation First, 500.0 is deposited so the balance is 2000. Then 1,000.0 is withdrawn. Finally, "invalid_type" is added as a transaction type, so the error is shown.